Workout 1A Homework 2 Challenge

Provide the best answer for each of the following questions.

Time Remaining 00:59:12

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

1

Review Later

If $\frac{8}{9} + \frac{10}{11} = 1 + \cfrac{1}{1 + \cfrac{1}{a + \cfrac{1}{b + \cfrac{1}{c}}}}$, where a, b, and c are integers, what are a, b, and c?

a =

b =

c =

2

Review Later

Which of the following is the formula for the sum of the first n squares, starting with $n = 1$?

- $\frac{n(n-1)(2n+1)}{6}$
- $\frac{n^2+n+1}{6}$
- $\frac{n(n+1)(n+2)}{6}$
- $\frac{n(n+1)(2n+1)}{6}$

3

Review Later

A recipe for puff pastry calls for rolling out the dough and folding it, done 6 times. How many layers are there in the puff pastry?

4

Review Later

What is 23564541 divided by 1001?

5

Review Later

Given that

$$(a+4)(a+3)(a-2)-(a+4)(a-6)=1400,$$

for integer a, find the value of a.

6

Review Later

Arnav, Bing and Chirag are playing a total of 30 card games, and bet $\$1$ on each game. The two losers pay $\$1$ each to the winner of the game. If Arnav ends up winning a net $\$15$, how many games did he win?

7

Review Later

Without using a calculator, what are the first 3 digits of $\frac{123456789}{1000001}$?

8

Review Later

Suppose

$$\begin{aligned} a+c+d &= -14 \\ b+c+d &= 6 \\ a+b+d &= -9 \\ a+b+c &= 5. \end{aligned}$$

What is the value of b?

9

Review Later

Arnav, Bing and Chirag are playing a total of 30 card games, and bet $\$1$ on each game. The two losers pay $\$1$ to the winner of each game. If Arnav ends up winning a net $\$15$ and Bing loses $\$9$ more than Chirag, then how many games did Chirag win?

10

Review Later

What is the sum $15+17+19+21+23+25+27+29+31+33+35$?

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11

Review Later

Arnav, Bing and Chirag are playing a total of 30 card games, and bet $\$1$ on each game. The two losers pay $\$1$ to the winner of each game. If Arnav ends up winning $\$6$ more than Bing and Bing wins $\$6$ more than Chirag, then how many games did Bing win?

12

Review Later

What is the sum of the following?

$$\begin{gathered}1001+1111+1221+1331+1441+1551+1661+1771+1881+1991+\\ 2002+2112+2222+2332+2442+2552+2662+2772+2882+2992+\\ 3003+3113+3223+3333+3443+3553+3663+3773+3883+3993+\\ 4004+4114+4224+4334+4444+4554+4664+4774+4884+4994+\\ \cdots\\ 9009+9119+9229+9339+9449+9559+9669+9779+9889+9999.\end{gathered}$$

Your answer should the number only, with no commas.

13

Review Later

What is the sum of the digits of the decimal that is equivalent to 5^{-6}?

14

Review Later

The first term of a sequence is equal to $(1007+2008+3009)$, the second term is equal to $(1014+2016+3018)$, the third term is equal to $(1021+2024+3027)$. What is the sum of

the first 14 terms of this sequence?

15

Review Later

Arnav, Bing and Chirag are playing a total of 30 card games, and bet $\$1$ on each game. The two losers pay $\$1$ to the winner of each game. If Bing ends up with a net positive total of money $\$21$ more than Chirag and Arnav ends up with a net $\$15$ more than Chirag, how many games did Arnav win?

16

Review Later

In Expiiland, a person's age is determined by their name. Each letter has a value, and the age of a person is the product of the letters in his or her name. A group of Expiidians, wanting to know the value of each letter, visit the town elder for help. The elder agrees to release the letter values, but only if the group can tell him the current age of the town founder, Sylbia. What number should the group give the elder? Your answer should the number only, without commas or units.

The Group of Expiidians

Name: Bill, Age: 10
Name: Lila, Age: 20
Name: Sally, Age: 30
Name: Ali, Age: 40

17

Review Later

In one beehive, honey is produced at a rate of 2 teaspoons per honeybee. In a second hive, honey is produced at a rate of 3 teaspoons per honeybee. Suppose that over the course of a month, the honey produced both hives was the same. What was the average number of

teaspoons of honey that one bee produced during this time? Your answer should be a simplified fraction in the form $\frac{a}{b}$, where a and b are integers.

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Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18

Review Later

An experimental aircraft is outfitted with both ramjet engines and normal jet engines, but can only use one or the other, not both at the same time. The ramjets accelerate by $21\frac{\text{mph}}{\text{sec}}$, and the normal jets accelerate the plane by $14\frac{\text{mph}}{\text{sec}}$. Suppose a plane's speed increase from ramjets is equal to its speed increase from normal jets. What was the average acceleration (rate of speed increase)? Your answer should be a decimal rounded to the nearest tenth, like 3.1 for π, for example.

19

Review Later

Without using a calculator, what are the first 3 digits of the product of 234523452345 and 428571428571?

20

Review Later

Suppose you have the digits 123456789. Put either a "$+$" or "$-$" between each digit. The result is called "S." Now add up all the possible **positive** "S" that we can achieve. What is this sum?

View Marked Questions

Finished

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